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Partial disassembly sequence planning based on cost-benefit analysis

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ABSTRACT

Due to growing concern for the environment, green design methods are used to create products that can reduce resource use and environmental impacts, throughout the entire product lifecycle. In particular, the end-of-life stage of the product lifecycle has received much attention recently. Disassembly sequence planning is one of the major product end-of-life methods, which can be used to increase the reuse rate of components and reduce product environmental impacts. However, total disassembly is usually expensive and impractical. Contrary to total disassembly, this study develops a new partial disassembly sequence planning method. Most prior partial disassembly sequence planning methods did not use life cycle impact assessment tools to perform cost-benefit analyses to find an optimized disassembly stopping point. In this study, Simapro Eco-indicator 99 is used to analyze environmental impacts. The proposed method considers part order, part disassembly directions, number of reorientations, and number of tool changes to find an optimized disassembly plan and an optimized disassembly stopping point. Study results show that the proposed partial disassembly sequence planning method can be used to reduce environmental costs and increase economical benefits, compared to the traditional disassembly methods. Study results also show that the method can help designers redesign the products, make the parts with high economical (recycling) benefits more accessible, and reduce disassembly cost.

1. Introduction

Due to growing concern for the environment, green design methods are becoming an important part of the design process. Basically, green design methods cover the entire product lifecycle, for example, design for assembly, supply chain management, lifecycle assessment, design for disassembly, design for remanufacture, and disassembly sequence planning. All of the methods have the same purpose: reduce environmental costs and increase economical benefits. In particular, the end-of-life (EOL) phase of the product lifecycle has received much attention recently (Gheorghe and Ishii, 2007). Many companies now are required, by law, to take back their EOL products for part repair, reuse, remanufacturing, and recycling to reduce the environmental impacts.

During the product EOL stage, products are generally first disassembled to recover high-value parts (Bogue, 2007). Therefore, disassembly serves as a basis for most of the EOL treatments, and it plays a key role in product lifecycle (Jovane et al., 1993). However, product disassembly is an expensive process, and it has high impacts on products' final EOL values. If a product is difficult to be disassembled, it might increase the cost to reuse or recycle.

Feldmann et al. (2001) mentioned that the number of disassembled parts also affects product EOL recovery profit. They suggested that an optimal disassembly sequence should be a balance between 'no disassembly' and 'complete disassembly' (as shown in Fig. 1). 'No disassembly' removes no parts, and the entire products are discarded when they reach their EOL stages, so 'no disassembly' increases the environmental impacts. 'Complete disassembly' removes all parts and recycle or reuse all parts, so 'complete disassembly' reduces the environmental impacts and increases recycling benefits, but it increases disassembly cost. Therefore, there exists an optimal EOL strategy, which places between 'no disassembly' and 'complete disassembly'.

In this research, sequential disassembly sequences are considered. In a sequential disassembly sequence, each disassembly level contains only one part. Disassembly process might stop at any disassembly level, which is called disassembly stopping point. An optimized disassembly stopping point (or disassembly level) between 'no disassembly' and 'complete disassembly' in a sequential disassembly sequence will be found. Therefore, in relation to the disassembly time in Fig. 1, the abscissa will be the disassembly stopping point.

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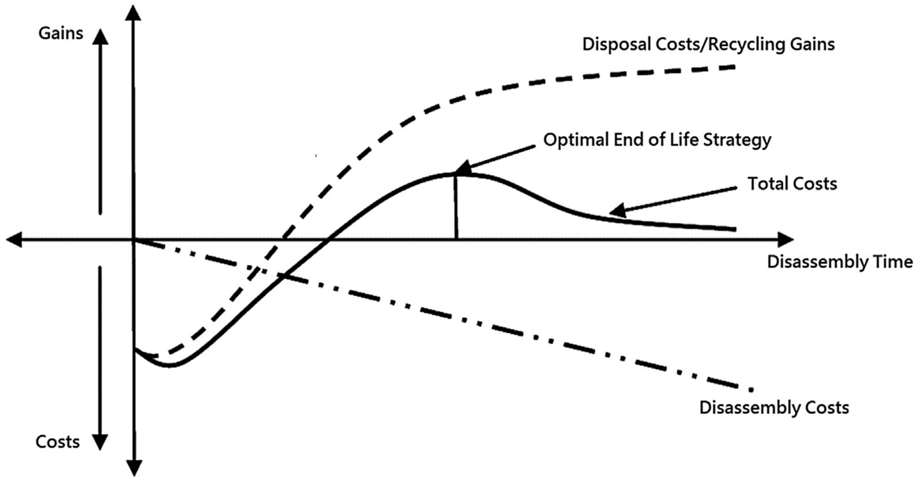

Fig. 1. Optimal end of life strategy (Feldmann et al., 2001).

2. Literature review

Smith and Hung (2015) gave a thorough review regarding different types of disassembly planning methods, e.g., total disassembly (or complete disassembly) and partial disassembly. Usually, total disassembly is expensive and impractical (Desai and Mital, 2003; Dong et al., 2007). Some methods use partial or selective disassembly to only remove parts that have higher environmental impacts or higher recycling values (Zhang et al., 2007; Behdad et al., 2010; Smith et al., 2012; Mitrouchev et al., 2015).

Some disassembly sequence planning methods use mathematical programming techniques to create optimal disassembly plans that minimize disassembly time. For example, integer programming techniques (Ma et al., 2011), dynamic linear programming techniques (Li et al., 2002), or mixed-integer programming techniques (Lambert, 2003; Behdad et al., 2014) are used to solve the problem. Some methods use heuristic searching techniques. For example, expert rules (Smith et al., 2012), heuristic algorithms (Shimizu et al., 2007; Xue et al., 2007; Wang and Shi, 2014), or case-based reasoning (Zeid et al., 1997; Veerakamolmal and Gupta, 2002) are used to simplify and facilitate the disassembly planning processes. However, most of the disassembly sequence planning methods did not use life cycle impact assessment tools, e.g., Eco-Indicator 99, to perform cost-benefit analyses or find an optimized disassembly stopping point.

Ma et al. (2011) used an extended AND/OR graph to generate all possible disassembly plans and used an integer programming model to conduct cost-benefit analysis. Achillas et al. (2013) performed cost-benefit analysis to determine the depth of disassembly for a given product. However, Achillas et al. did not consider disassembly sequence planning or environmental impacts in their cost-benefit analysis. Riggs et al. (2015) presented a two-stage sequence model for product disassembly. They pushed the components with high reuse value to the front of the disassembly sequence. Riggs et al. did not consider the environmental impacts or the disassembly cost in their mixed integer program.

This study develops a new disassembly sequence planning method by cost-benefit and rule-based analysis. It would be very time consuming and physically impractical to find a global optimum solution. This study, an optimized sequential disassembly plan is generated based on given expert rules. An optimized disassembly stopping point is then selected using cost-benefit analysis. Simapro Eco-indicator 99 is used to analyze environmental impacts. The developed disassembly sequence planning method can create disassembly plans that are functionally, physically, environmentally, and economically practical.

3. Disassembly sequence planning

3.1. Product model

In this study, a product model is represented by constraint matrices. Each row in a constraint matrix represents a part. Each column in a constraint matrix represents a constraint. Each row and column element in a constraint matrix contains a constraint for a part. A constraint can be functional, physical, environmental, or economical. A functional constraint describes the function of a part. A physical constraint is a contact or motion constraint. An environmental constraint is an environmental cost constraint. An economical constraint is a benefit constraint. A part is a component or fastener.

As a result, a design is separated into a functional constraint matrix for components, a physical constraint matrix for components, an environmental constraint matrix for components, an economical constraint matrix for components, a functional constraint matrix for fasteners, a physical constraint matrix for fasteners, an environmental constraint matrix for fasteners, and an economical constraint matrix for fasteners. A physical constraint matrix is separated into a contact constraint matrix and a motion constraint matrix.

3.2. A disassembly plan

A disassembly plan is an ordered set of parts and disassembly directions. Parts are components and fasteners. An ordered set of parts optimizes part order (PO), part disassembly directions (PD), number of removed parts (NP), number of reorientations (NR), and number of tool changes (NT), based upon functional, physical, environmental, and economical constraints. Disassembly directions include $\{+x, -x, +y, -y, +z, -z\}$ directions.

3.3. Expert rules and cost-benefit analysis

Instead of generating all possible disassembly plans, to facilitate disassembly sequence planning, in this study, expert rules are used to find an optimized sequential disassembly plan that removes all parts, based upon environmental costs and economical benefits (Rules 1–5). Cost-benefit analysis is then used to select an optimized stopping point (or disassembly level) for the disassembly plan (Rules 6–8).

Rule 1: Remove the part that has the highest economical benefit first.
Rule 2: Remove the parts that can be removed in the same disassembly direction before removing the parts that can be removed in different disassembly directions.
Rule 3: Remove the parts that use the same tool before removing the parts that use different tools.
Rule 4: Change disassembly directions by 90° before changing disassembly directions by 180°.
Rule 5: Reorder the parts to create disassembly plans that are optimized for the number of tool changes.
Rule 6: Select the stopping point that has the maximum (total benefits - total costs).
Rule 7: Select the stopping points that have at least 85% of the maximum (total benefits–total costs) and at most 70% of the corresponding environmental impacts.
Rule 8: If there is more than one stopping point in Rule 7, select the stopping point with the highest (total benefits–total costs) in Rule 7.

Algorithm 1 in Fig. 2 shows steps for creating an optimized total disassembly sequence plan. Step 1 uses Rule 1 to select the first disassembly direction and disassembly tool. Steps 2 and 3 use Rules 2 and 3 to remove parts from the same direction and parts using the same tool. Step 6 uses Rule 4 to change disassembly directions by 90°, before changing disassembly directions by 180°. Finally, Step 17 uses Rule 5 to reorder the parts to create a disassembly plan that is optimized for the number of tool changes.

Algorithm 2 in Fig. 3 shows steps for selecting an optimized disassembly stopping point. Step 1 uses Rule 6 to select a stopping point that creates an optimized disassembly plan with maximum (total benefits–total costs). Step 2 uses Rule 7 to select stopping points that have at least 85% of the maximum (total benefits–total costs) and at most 70% of the corresponding environmental impacts. If there exists a stopping point in Step 2, Step 4 uses Rule 8 to return the stopping point with the highest (total benefits–total costs). Otherwise, return the stopping point found in Step 1. Algorithms 1 and 2 are fully automated without manual jobs. The only manual jobs are the preprocessing work of the constraint matrices.

3.4. The environmental costs

For a disassembly plan with (n) parts, (p) parts that are removed and repaired, replaced, or recycled, and (n–p) parts that are not removed and shredded, the environmental costs (EC_i) of the ($i = 1 \ldots n$) parts are the disassembly costs (DC_i), sorting costs (OC_i), and repair, replacement, or recycling costs (RC_i) of the ($i = 1 \ldots p$) parts that are removed and repaired, replaced, or recycled, and the environmental impacts (EI_i), and shredding costs (SC_i) of the ($i = p{+}1 \ldots n$) parts that are not removed and shredded.

Desai and Mital (2003) considered several disassembly factors to provide a scoring system to evaluate the ease of disassembly for part i (ED_i). The higher the score, the more difficult of the disassembly process is. The disassembly factors include part accessibilities (A_i), part positions (P_i), disassembly tools (T_i), disassembly forces (F_i), and material handling costs (M_i) (in standard units):

$$ED_i = (A_i + P_i + T_i + F_i + M_i) \tag{1}$$

The values of the variables in standard units are the weights of

Step	**Algorithm 1: Disassembly Sequence Planning**
1	Select a part to be disassembled using Rule 1; $d \leftarrow$ disassembly direction; $t \leftarrow$ disassembly tool;
2	FOR (all parts which can be disassembled in direction d) DO
3	FOR (all parts p which can be disassembled using tool t) DO
4	Disassemble part p using tool t in direction d;
5	$t \leftarrow$ Change to a new disassembly tool;
6	Set $D \leftarrow$ Directions ($d \pm 90°$) in which parts can be disassembled;
7	IF $D \neq \Phi$ THEN
8	Subset $D_T \leftarrow$ Directions in Set D in which parts can be disassembled using tool t;
9	IF $D_T \neq \Phi$ THEN
10	$d \leftarrow$ Change to a new disassembly direction in Subset D_T using Rule 1;
11	Go to Step 2;
12	ELSE $d \leftarrow$ Change to a new disassembly direction in Set D using Rule 1;
13	Go to Step 2;
14	ELSE Set $D \leftarrow$ Directions ($d \pm 180°$) in which parts can be disassembled;
15	IF $D \neq \Phi$ THEN
16	Go to Step 8;
17	Rule 5;
18	END

Fig. 2. Algorithm 1 for creating an optimized disassembly sequence plan.

Step	**Algorithm 2: Stopping Point Selection**
1	p ← Stopping point that creates an optimized disassembly plan with maximum (total benefits - total costs);
2	Set S ← Stopping points that have at least 85% of the maximum (total benefits - total costs) and at most 70% of the corresponding environmental impacts;
3	IF $S \neq \Phi$ THEN
4	p ←Select the stopping point with the highest (total benefits - total costs) in S.
5	RETURN p;
6	END

Fig. 3. Algorithm 2 for selecting an optimized disassembly stopping point.

the variables, and they are unitless scores. Based on the evaluation scores given by Desai and Mital (2003), this study adds part orders (PO_i), part disassembly directions (PD_i), number of removed parts (NP_i), number of reorientations (NR_i), and number of tool changes (NT_i) into disassemblability evaluation (in standard units):

$$DC_i = (ED_i + PO_i + PD_i + NP_i + NR_i + NT_i) \quad (2)$$

Although different variables represent different disassembly factors, the summation of the weights represents a compound score of the disassemblability of a part. The higher the score, the more difficult to disassemble part *i*. A conversion factor ((100/180) standard units/degrees of rotation) can be used to convert the number of reorientations (NR_i) in degrees into standard units, and a conversion factor (10 standard units/number of tool changes) can be used to convert the number of tool changes (NT_i) into standard units.

A conversion factor (0.36 s/standard unit) can be used to convert the disassembly costs (DC_i) in standard units into disassembly times (DT_i) in seconds, and a conversion factor (0.055 dollars/second) can be used to convert the disassembly times (DT_i) in seconds into disassembly costs (DC_i) in dollars.

The sorting costs (OC_i) in dollars can be measured or estimated from the labor costs, sorting fees, or energy costs in dollars. The repair, replacement, or recycling costs (RC_i) in dollars can be measured or estimated from the repair, replacement, or recycling costs per weight (RC_i) (in dollars/gram):

$$RC_i \text{ (in dollars)} = RC_i \text{ (in dollars/gram)} \cdot W_i \text{ (in grams)} \quad (3)$$

The environmental impacts (EI_i) in milli-points can be estimated from the environmental impacts per weight (in Eco-Indicator 99 milli-points/gram):

$$EI_i \text{ (in milli-points)} = EI_i \text{ (in milli-points/gram)} \cdot W_i \text{ (in grams)} \quad (4)$$

A conversion factor (0.0017 dollars/milli-point) can be used to convert the environmental impacts (EI_i) in milli-points into environmental impacts (EI_i) in dollars. The conversion factor (0.0017 dollars/milli-point) is the product of the estimated environmental impact of one person per year (1 ha) and the estimated price of 1 ha of land (1700 dollars/hectare), divided by the estimated environmental impact of one person per year (1,000,000 milli-points).

The shredding costs (SC_i) in dollars can be estimated from the estimated shredding costs per weight (SC_i) (in dollars/gram):

$$SC_i \text{ (in dollars)} = SC_i \text{ (in dollars/gram)} \cdot W_i \text{ (in grams)} \quad (5)$$

3.5. The economical benefits

For a disassembly plan with (n) parts, (p) parts that are removed and repaired, replaced, or recycled, and (n - p) parts that are not removed and shredded, the economical benefits (EB_i) of the ($i = 1 \ldots n$) parts are the repair, replacement, or recycling benefits (RB_i) of the ($i = 1 \ldots p$) parts, and the shredding benefits (SB_i) of the ($i = p + 1 \ldots n$) parts that are not removed and shredded.

The repair, replacement, or recycling benefits (RB_i) in dollars can be estimated from the benefits per weight (RB_i) (in dollars/gram):

$$RB_i \text{ (in dollars)} = RB_i \text{ (in dollars/gram)} \cdot W_i \text{ (in grams)} \quad (6)$$

The shredding benefits (SB_i) in dollars can be estimated from the shredding benefits per weight (SB_i) (in dollars/gram):

$$SB_i \text{ (in dollars)} = SB_i \text{ (in dollars/gram)} \cdot W_i \text{ (in grams)} \quad (7)$$

3.6. The optimized stopping point

The optimized stopping point is generally the point at which the total cost-benefit (TCB) of the disassembly plan is optimized. However, the optimized stopping point is also the point at which the environmental impact of the disassembly plan is optimized.

The total cost-benefit (TCB) of the disassembly plans is the difference between the environmental cost (EC) of the disassembly plan and the economical benefit (EB) of the disassembly plan:

$$TCB = EC - EB \quad (8)$$

The environmental cost (EC) of the disassembly plan is equal to the sum of the environmental costs of the ($i = 1 \ldots p$) parts that are removed and repaired, replaced, or recycled and the ($i = p + 1 \ldots n$) parts that are not removed and shredded:

$$EC = \sum_{i=1}^{p}(DC_i + OC_i + RC_i) + \sum_{i=p+1}^{n}(EI_i + SC_i) \quad (9)$$

The economical benefit (EB) of the disassembly plan is equal to the sum of the environmental benefits of the ($i = 1 \ldots p$) parts that are removed and repaired, replaced, or recycled and the ($i = p+1 \ldots n$) parts that are not removed and shredded:

$$EB = \sum_{i=1}^{p}(RB_i) + \sum_{i=p+1}^{n}(SB_i) \quad (10)$$

4. Case studies

4.1. Case study 1 - the drive assembly

Fig. 4 shows Case study 1 - the drive assembly (Boothroyd, 1994; Smith et al., 2012). Simapro Eco-indicator 99 is used to analyze the environmental costs and economical benefits. A custom program written in Visual Basic is used to analyze the constraint matrices and create a complete disassembly plan, and select an optimized stopping point. Users need to manually input the functional, physical, environmental, and economical constraint matrices. The program will automatically generate an optimized disassembly plan and an optimized disassembly stopping point.

Table 1 shows the functional constraint matrix for components. Each row represents a component (1–10). Each row and column element contains the function of a component. For example, the function of component 1 is a 'base'. Table 2 shows the contact constraint matrix for components. Each row represents a component (1–10). Each column represents a disassembly direction (+x, −x, +y, −y, +z, −z). Each row and column element contains a (0), or the components (1–10) that contact a component in a given disassembly direction.

Table 3 shows the motion constraint matrix for components. Each row represents a component (1–10). Each column represents a disassembly direction (+x, −x, +y, −y, +z, −z). Each row and column element contains a (0), or the components (1–10) that intersect a component's projection in a given disassembly direction, and the fasteners (A-I) that connect a component to other components in a given disassembly direction.

Table 4 shows the environmental constraint matrix for components. Each row represents a component (1–10). Each column represents an environmental constraint (disassembly tool, ease of disassembly, component material, or component weight). Each row and column element contains an environmental constraint (disassembly tool, ease of disassembly, component material, or component weight) for a component.

Table 5 shows the economical constraint matrix for components. Each row represents a component (1–10). Each column represents an economical constraint (recycling benefit). Each row and column element contains an economical constraint (recycling benefit) for a component.

Table 1
Case study 1 - the functional constraint matrix for components.

Component	Function
1	Base
2	Spacer
3	Spacer
4	Cover
5	Control
6	Electromotor
7	Shell
8	Sensor
9	Spacer
10	Spacer

Table 6 shows the functional constraint matrix for fasteners. Each row represents a fastener (A-I). Each row and column element contains the function of a fastener.

Table 7 shows the contact constraint matrix for fasteners. Each row represents a fastener (A-I). Each column represents a disassembly direction (+x, −x, +y, −y, +z, −z). Each row and column element contains a (−1) for a fastener that cannot be removed in a disassembly direction, and a (0) for a fastener that can be removed in a disassembly direction.

Table 8 shows the motion constraint matrix for fasteners. Each row represents a fastener (A-I). Each column represents a disassembly direction (+x, −x, +y, −y, +z, −z). Each row and column element contains a (0), or the components (1–10) that intersect a fastener's projection in a given disassembly direction.

Table 9 shows the environmental constraint matrix for fasteners. Each row represents a fastener (A-I). Each column represents an environmental constraint (disassembly tool, ease of disassembly, fastener material, or fastener weight). Each row and column element contains an environmental constraint (disassembly tool, ease of disassembly, fastener material, or fastener weight) for a fastener.

Table 10 shows the economical constraint matrix for fasteners. Each row represents a fastener (A-I). Each column represents an economical constraint (recycling benefit). Each row and column element contains an economical constraint (recycling benefit) for a fastener.

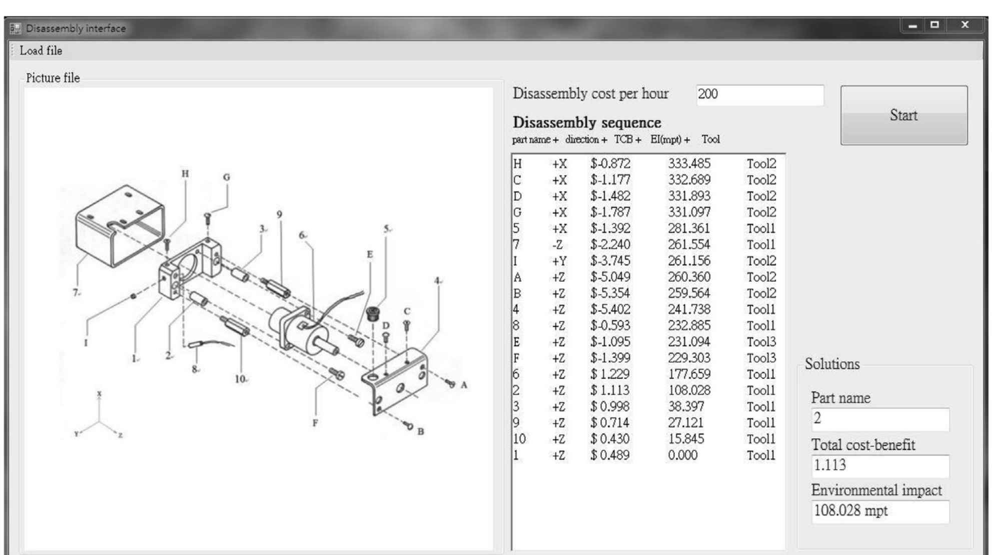

Fig. 4. Case study 1 - the drive assembly.

Table 2
Case study 1 - the contact constraint matrix for components.

Component	+x	−x	+y	−y	+z	−z
1	2,3,6,7,8,9,10	2,3,6,7,8,9,10	2,3,6,7,8,9,10	2,3,6,7,8,9,10	2,3,6,8,9,10	0
2	1	1	1	1	0	1
3	1	1	1	1	0	1
4	5.,6,7	6,7	5,6,7	5,6,7	5	5,6,9,10
5	0	4	4	4	4	4
6	4,7	4,7	1,4,7	1,4,7	4	1
7	1,4,5,6	1,4,6	1,.4,5	1,4,5	5	0
8	1	1	1	1	0	1
9	1	1	1	1	4	1
10	1	1	1	1	4	1

Table 3
Case study 1 - the motion constraint matrix for components.

Component	+x	−x	+y	−y	+z	−z
1	E,F,G,H,I	E,F,G,H,I	E,F,G,H,I	E,F,G,H,I	4,E,F,G,H,I	7,E,F,G,H,I
2	8	10	7	6	4	7
3	9	7	6	7	4	7
4	5,A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D	2,3,A,B,C,D
5	0	4	0	6	0	1
6	E,F	E,F	2,10,E,F	3,9,E,F	E,F	7,E,F
7	8,C,D,G,H	9,C,D,G,H	3,C,D,G,H	2,C,D,G,H	6,C,D,G,H	C,D,G,H
8	7	2	7	6	4	7
9	7,A	3,A	6,A	7,A	A	7,A
10	2,B	7,B	7,B	6,B	B	7,B

Table 4
Case study 1 - the environmental constraint matrix for components.

Component	Disassembly tool (in tool number)	Ease of disassembly (in standard units)	Component material (in material)	Component weight (in grams)
1	1	13.400	Abs	200
2	1	12.600	Steel	350
3	1	12.600	Steel	350
4	1	10.800	Abs	225
5	1	12.000	Steel	250
6	1	13.000	Aluminum	700
7	1	12.800	Abs	250
8	1	10.300	Aluminum	120
9	1	16.900	Abs	250
10	1	16.900	Abs	250

Table 5
Case study 1 - the economical constraint matrix for components.

Component	Recycling benefit (in dollars)
1	0.300
2	0.018
3	0.018
4	0.338
5	0.750
6	3.000
7	0.375
8	5.000
9	0.035
10	0.035

Table 6
Case study 1 - the functional constraint matrix for fasteners.

Fastener	Function
A	Fastener
B	Fastener
C	Fastener
D	Fastener
E	Fastener
F	Fastener
G	Fastener
H	Fastener
I	Fastener

Table 11 shows the complete disassembly plan created by Rules 1–4. The disassembly plan removes Component 5 first because Component 5 has a high economical (recycling) benefit. The disassembly plan removes nineteen parts, with three reorientations, and six tool changes.

Table 12 shows the complete disassembly plan, reordered by Rule 5. The disassembly plan removes Fasteners H-C-D-G first, to reduce tool changes. The disassembly plan removes nineteen parts, with three reorientations, and five tool changes.

Table 13 and Fig. 5 show the total disassembly cost and total recycling benefit of the ($i = 1 \ldots p$) parts that are removed and repaired, replaced, and recycled, the total environmental impacts of the ($i = p+1 \ldots n$) parts that are not removed and shredded, and the (total benefit - total cost) of the disassembly plan (in dollars), as functions of the stopping point (p). Fig. 6 shows the total environmental impacts of the ($i = p+1 \ldots n$) parts that are not removed and shredded (in milli-points), as a function of the stopping point

Table 7
Case study 1 - the contact constraint matrix for fasteners.

Fasteners	+x	−x	+y	−y	+z	−z
A	−1	−1	−1	−1	0	−1
B	−1	−1	−1	−1	0	−1
C	0	−1	−1	−1	−1	−1
D	0	−1	−1	−1	−1	−1
E	−1	−1	−1	−1	0	−1
F	−1	−1	−1	−1	0	−1
G	0	−1	−1	−1	−1	−1
H	0	−1	−1	−1	−1	−1
I	−1	−1	0	−1	−1	−1

Table 8
Case study 1 - the motion constraint matrix for fasteners.

Fasteners	+x	−x	+y	−y	+z	−z
A	0	0	0	0	0	0
B	0	0	0	0	0	0
C	0	0	0	0	0	0
D	0	0	0	0	0	0
E	0	0	0	0	4	0
F	0	0	0	0	4	0
G	0	0	0	0	0	0
H	0	0	0	0	0	0
I	0	0	7	0	0	0

Table 9
Case study 1 - the environmental constraint matrix for fasteners.

Parts	Disassembly tool (in tool number)	Ease of disassembly (in standard units)	Fastener material (in material)	Fastener weight (in grams)
A	2	15.300	Steel	4
B	2	15.300	Steel	4
C	2	15.300	Steel	4
D	2	15.300	Steel	4
E	3	15.300	Steel	9
F	3	15.300	Steel	9
G	2	15.300	Steel	4
H	2	15.300	Steel	4
I	2	15.300	Steel	2

Table 10
Case study 1 - the economical constraint matrix for fasteners.

Fastener	Recycling benefit (in dollars)
A	0.000
B	0.000
C	0.000
D	0.000
E	0.000
F	0.000
G	0.000
H	0.000
I	0.000

(p). The total disassembly cost, total recycling benefit, total environmental impacts, and (total benefit - total cost) of the disassembly plan (in dollars) show that none of the parts, all of the parts, all of the parts, and (14) of the parts should be removed and repaired, replaced, or recycled. The total environmental impacts of the disassembly plan (in milli-points) in Fig. 6 shows that all of the parts should be removed and repaired, replaced, or recycled.

When (p) is (14) (before removing Part 2), the total disassembly cost, total recycling benefit, total environmental impacts, and (total benefit - total cost) of the disassembly plan are ($7.932), ($9.463), (177.659 mpt), and ($1.229). Algorithm 2 is used to select an optimized stopping point for which:

$$(\text{total benefit} - \text{total cost}) \geq (0.85) \cdot (1.229)\ (\text{in dollars}) \tag{11}$$

$$(\text{environmental impacts}) \leq (0.70) \cdot (177.659)\ (\text{in mpt}) \tag{12}$$

$$(\text{total benefit} - \text{total cost}) \geq (1.045)\ (\text{in dollars}) \tag{13}$$

$$(\text{environmental impacts}) \leq (124.36)\ (\text{in mpt}) \tag{14}$$

Therefore, the new optimized stopping point is (p) = (15) (after removing part 2), for which the total disassembly cost, total recycling benefit, total environmental impacts, and (total benefit−total cost) of the disassembly plan are ($8.184), ($9.481), (108.028 mpt), and ($1.113).

The total disassembly cost, total recycling benefit, total environmental impacts, and (total benefit - total cost) for removing none of the parts are ($0.000), ($0.000), (334.281 mpt), and ($−0.568). The total disassembly cost, total recycling benefit, total environmental impacts, and (total benefit - total cost) for removing

Table 11
Case study 1 - the complete disassembly plan created by Algorithm 1.

Part order	Part	Disassembly direction	Disassembly tool
1	5	+x	1
2	C	+x	2
3	D	+x	2
4	G	+x	2
5	H	+x	2
6	7	−z	1
7	I	+y	2
8	A	+z	2
9	B	+z	2
10	4	+z	1
11	8	+z	1
12	E	+z	3
13	F	+z	3
14	6	+z	1
15	2	+z	1
16	3	+z	1
17	9	+z	1
18	10	+z	1
19	1	+z	1

Table 12
Case study 1 - the complete disassembly plan, reordered by Rule 5.

Part order	Part	Disassembly direction	Disassembly tool
1	H	+x	2
2	C	+x	2
3	D	+x	2
4	G	+x	2
5	5	+x	1
6	7	−z	1
7	I	+y	2
8	A	+z	2
9	B	+z	2
10	4	+z	1
11	8	+z	1
12	E	+z	3
13	F	+z	3
14	6	+z	1
15	2	+z	1
16	3	+z	1
17	9	+z	1
18	10	+z	1
19	1	+z	1

Table 13
Case study 1 - the cost-benefit analysis of the disassembly plan.

Stopping point (p)	Part	Total disassembly cost (in dollars)	Total recycling benefit (in dollars)	Total environmental impacts (in dollars)	Total benefit−total cost (in dollars)
0	–	0.000	0.000	0.568	−0.568
1	H	0.306	0.000	0.567	−0.873
2	C	0.612	0.000	0.566	−1.178
3	D	0.918	0.000	0.564	−1.482
4	G	1.224	0.000	0.563	−1.787
5	5	1.664	0.750	0.478	−1.392
6	7	2.920	1.125	0.445	−2.240
7	I	4.426	1.125	0.444	−3.745
8	A	5.732	1.125	0.443	−5.050
9	B	6.038	1.125	0.441	−5.354
10	4	6.454	1.463	0.411	−5.402
11	8	6.660	6.463	0.396	−0.593
12	E	7.166	6.463	0.393	−1.096
13	F	7.472	6.463	0.390	−1.399
14	6	7.932	9.463	0.302	1.229
15	2	8.184	9.481	0.184	1.113
16	3	8.436	9.499	0.065	0.998
17	9	8.774	9.534	0.046	0.714
18	10	9.112	9.569	0.027	0.430
19	1	9.380	9.869	0.000	0.489

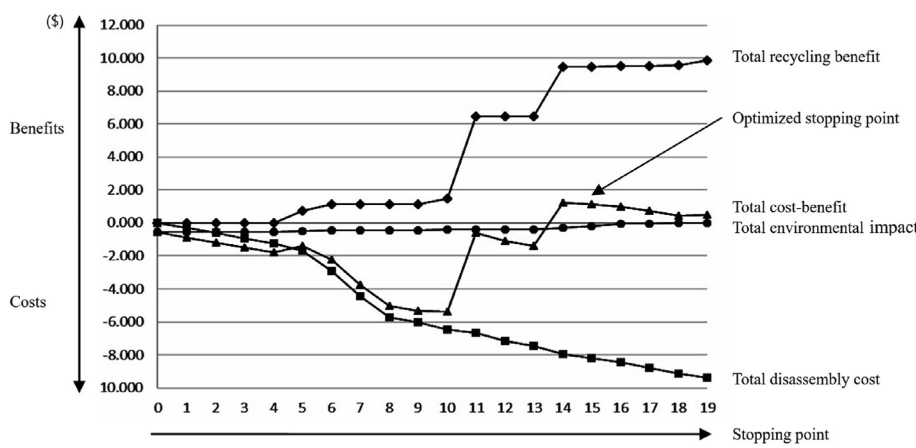

Fig. 5. Case study 1 - the cost-benefit analysis of the disassembly plan.

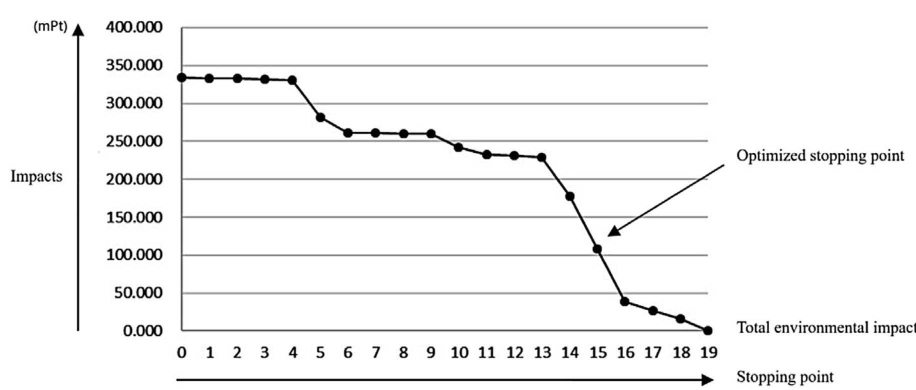

Fig. 6. Case study 1 - the environmental impact of the disassembly plan.

all of the parts are ($9.380), ($9.869), (0.000 mpt), and ($0.489). The cost-benefit analysis shows that the new green design methods, which remove the optimized number of parts, can be used to reduce environmental costs and increase economical benefits by (|(1.113)−(0.489)|)/|(0.489)| = (127.61) percent, compared to removing all of the parts.

4.2. Case study 2 - the Epson stylus scan 2500 multi-function scanner assembly

Fig. 7 shows Case study 2 - the Epson Stylus Scan 2500 multi-function scanner assembly (Epson, 2013). Epson Stylus Scan 2500 multi-function scanner assembly is an existing product. The design

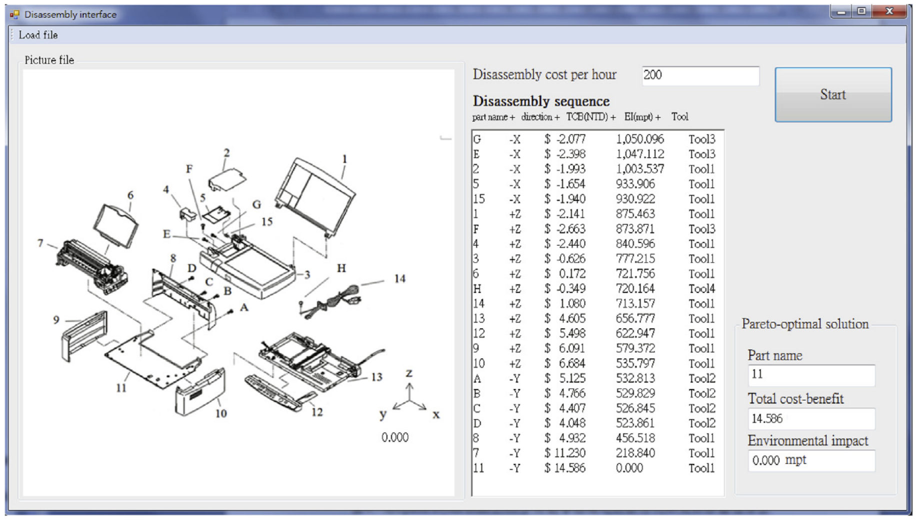

Fig. 7. Case study 2 - the Epson Stylus Scan 2500 multi-function scanner assembly.

Table 14
Case study 2 - the complete disassembly plan.

Part order	Part	Disassembly direction	Disassembly tool
1	G	−x	3
2	E	−x	3
3	2	−x	1
4	5	−x	1
5	15	−x	1
6	1	+z	1
7	F	+z	3
8	4	+z	1
9	3	+z	1
10	6	+z	1
11	H	+z	4
12	14	+z	1
13	13	+z	1
14	12	+z	1
15	9	+z	1
16	10	+z	1
17	A	−y	2
18	B	−y	2
19	C	−y	2
20	D	−y	2
21	8	−y	1
22	7	−y	1
23	11	−y	1

for the Epson Stylus Scan 2500 multi-function scanner assembly does not include constraint matrices. As a result, constraint matrices are derived from the design. A custom program is used to analyze the constraint matrices and create a complete disassembly plan. Simapro Eco-indicator 99 is used to analyze the environmental costs and economical benefits. Table 14 shows the complete disassembly plan, created by Algorithm 1. The disassembly plan removes twenty-three parts, with two reorientations, and seven tool changes.

Figs. 8–9 show the cost-benefit analysis of the disassembly plan. The cost-benefit analysis shows that the optimized stopping point is (p) = (23) (after removing all of the parts), for which the (total benefit−total cost) of the disassembly plan is $14.586, and the total environmental impact of the disassembly plan is 0.00 mpt. The cost-benefit analysis shows that all of the parts must be removed. As a result, the cost-benefit analysis shows that the product could be redesigned to make the parts with high economical (recycling) benefits more accessible, to reduce disassembly cost.

The disassembly plan created by the new green design method is compared with the disassembly plan recommended by Epson (Epson, 2013), which is {A, B, C, D, 8, 1, 2, 4, 5, 15, E, G, F, 3, 12, 9, 10, 13, 6, 7, H, 14, 11}. The disassembly plan removes twenty-three parts, with nine reorientations, and five tool changes. Since each

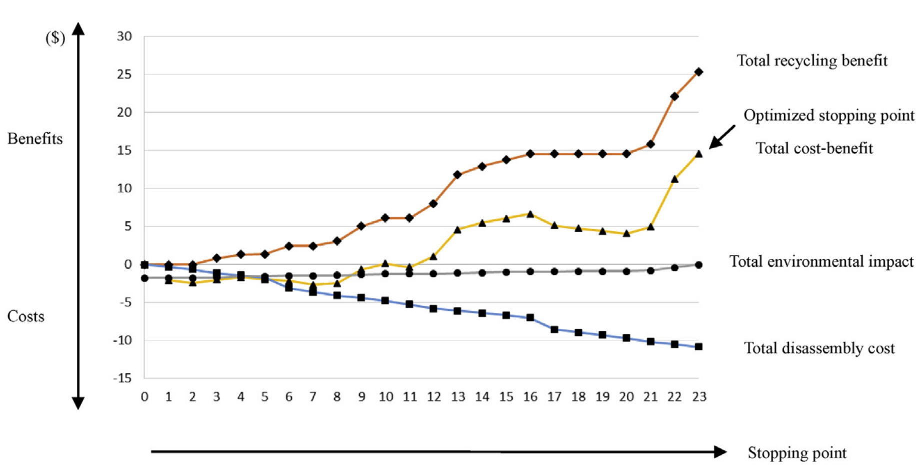

Fig. 8. Case study 2 - the cost-benefit analysis of the disassembly plan.

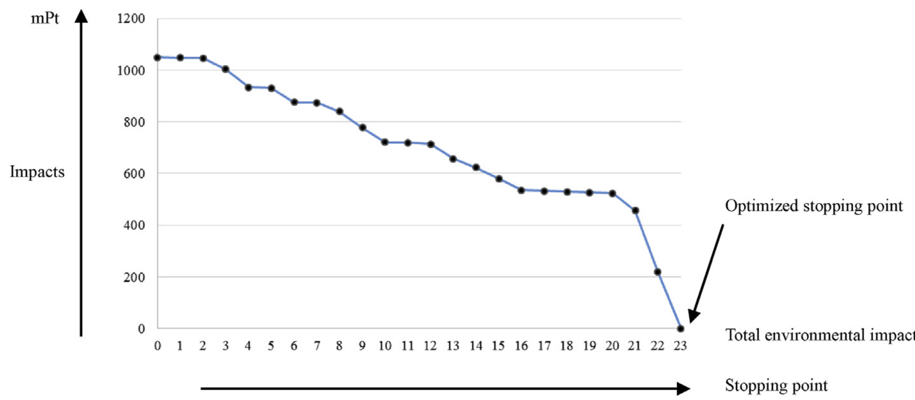

Fig. 9. Case study 2 - the environmental impact of the disassembly plan.

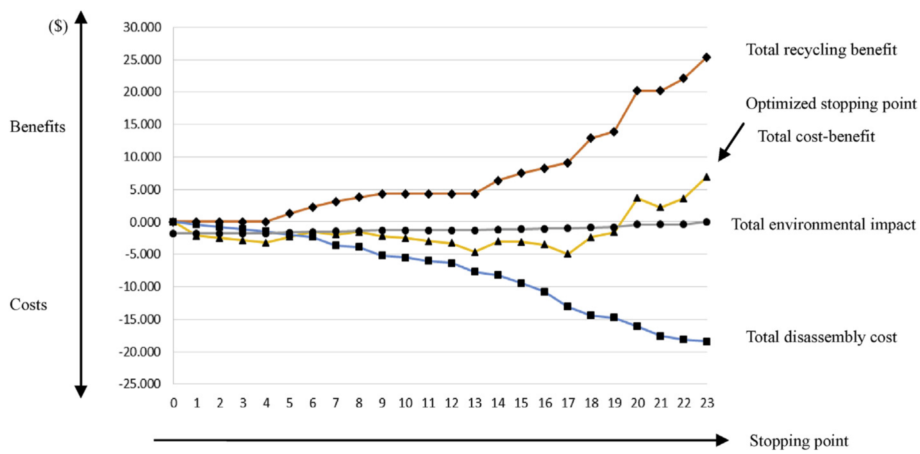

Fig. 10. Case study 2 - the cost-benefit analysis of the disassembly plan recommended by Epson (2013).

reorientation needs refixturing, it costs much more than tool changes. Therefore, although the disassembly plan recommended by Epson has fewer tool changes, more reorientations lower the total cost benefit. Figs. 10 and 11 show the cost-benefit analysis of the disassembly plan recommended by Epson. The cost-benefit analysis shows that the optimized stopping point is also (p) = (23) (after removing all of the parts). However, the (total benefit - total cost) of the disassembly plan is (6.994), which is lower than the result (14.586) obtained by the new green design method. Although both methods remove all of the parts for this case study, the cost-benefit analysis shows that the new green design method can be used to reduce environmental costs and increase economical benefits by $(|(14.586)-(6.994)|)/|(6.994)| = (108.55)$ percent, compared to the traditional method.

5. Conclusions

This study proposes a new partial disassembly sequence

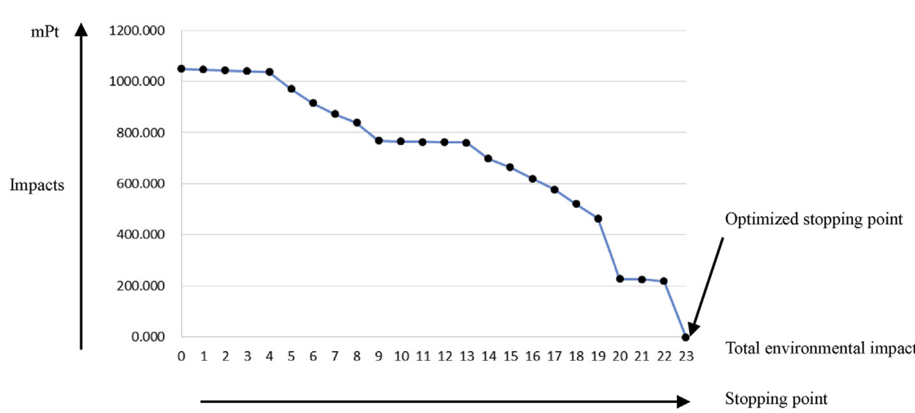

Fig. 11. Case study 2 - the environmental impact of the disassembly plan recommended by Epson (2013).

planning by cost-benefit analysis. The aim of this research is to find an optimized disassembly sequence and disassembly stopping point, which maximizes economical benefits and minimizes environmental costs. During the design process, the method uses constraint matrices and expert rules to create disassembly models and disassembly plans that are functionally, physically, environmentally, and economically practical.

Designers can use the proposed new green design method to improve designs during the design process, and remove an optimized number of parts during the reuse and replacement processes. Case study 1 shows that the method can be used to reduce environmental costs and increase economical benefits by 127.61 percent, compared to removing all of the parts. Case study 2 shows that although both the new green design method and the traditional method remove all of the parts, the new green design method, which optimizes part order, part disassembly directions, number of reorientations, and number of tool changes, can reduce environmental costs and increase economical benefits by 108.55 percent, compared to the traditional method. Case study 2 also shows that the product could be redesigned to make the parts with high economical (recycling) benefits more accessible, to reduce the disassembly cost.

The new disassembly sequence planning method can be improved by adding new features. For example, the method can be improved by adding new features for parts that are removed and scrapped. The method can also be used to reduce environmental costs and increase economical benefits for other products, and for other functional, environmental, economical, or physical constraints.

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